

§2.6

Order of Operations

One agreed order, so that every expression has exactly one value

Today's Goal

LT.8 Order of Operations

I can apply the order of operations correctly to evaluate a numerical expression.

- Explain why we need one agreed-upon convention for evaluating expressions
- Apply the order of operations, including nested grouping symbols
- Work same-tier operations from left to right

Four Students, Four Answers

Four students evaluate $12 + 24 \div 6 \times 2$. What did each one do first? Which is right?

Student 1.

$$\begin{aligned} & 36 \div 6 \times 2 \\ & = 6 \times 2 = 12 \end{aligned}$$

Student 2.

$$\begin{aligned} & 36 \div 6 \times 2 \\ & = 36 \div 12 = 3 \end{aligned}$$

Student 3.

$$\begin{aligned} & 12 + 24 \div 12 \\ & = 12 + 2 = 14 \end{aligned}$$

Student 4.

$$\begin{aligned} & 12 + 4 \times 2 \\ & = 12 + 8 = 20 \end{aligned}$$

One Agreed Order

Definition 2.6.1

The *order of operations* is the agreed-upon convention for the sequence in which the operations in a numerical expression are evaluated, so that every expression has a single, unambiguous value.

1. *Grouping symbols.* Evaluate anything inside grouping symbols (parentheses, brackets) first. When grouping symbols are nested within one another, start with the innermost and work outward.
2. *Exponents.* Evaluate powers.

One Agreed Order, Continued

Definition 2.6.1, continued

3. *Multiplication and division.* Perform multiplications and divisions as they appear from left to right.
4. *Addition and subtraction.* Perform additions and subtractions as they appear from left to right.

PEMDAS and GEMDAS

PEMDAS

Parentheses

Exponents

Multiplication and **D**ivision (left to right)

Addition and **S**ubtraction (left to right)

GEMDAS

Grouping Symbols

Exponents

Multiplication and **D**ivision (left to right)

Addition and **S**ubtraction (left to right)

GEMDAS reminds us that *any* grouping symbol comes first, not just parentheses.

The Left-to-Right Trap

Watch out

Multiplication does not always come before division, and addition does not always come before subtraction.

$$\begin{aligned} & 12 + 24 \div 6 \times 2 \\ = & 12 + 4 \times 2 && \text{division first, it is leftmost} \\ = & 12 + 8 && \text{then multiplication} \\ = & 20 \end{aligned}$$

Student 4 is right.

Working From the Inside Out

Evaluate $3 \times (8 - [10 - (4 + 3)])$. Start from the inside and work out.

$$\begin{aligned} & 3 \times (8 - [10 - (4 + 3)]) \\ = & 3 \times (8 - [10 - 7]) && \text{innermost group: } (4 + 3) \\ = & 3 \times (8 - 3) && \text{next group: } [10 - 7] \\ = & 3 \times 5 && \text{last group: } (8 - 3) \\ = & 15 \end{aligned}$$

No Grouping Symbols At All

Evaluate $24 \div 6 \times 3 + 5$.

$$\begin{aligned} & 24 \div 6 \times 3 + 5 \\ &= 4 \times 3 + 5 && \text{division first, it is leftmost} \\ &= 12 + 5 && \text{then multiplication} \\ &= 17 \end{aligned}$$

Two Groups, Side by Side

Evaluate $(2 + 6) \times [3^2 - 15 \div 5]^2 + 4$.

$$\begin{aligned} & (2 + 6) \times [3^2 - 15 \div 5]^2 + 4 \\ = & 8 \times [3^2 - 15 \div 5]^2 + 4 \\ = & 8 \times [9 - 15 \div 5]^2 + 4 \\ = & 8 \times [9 - 3]^2 + 4 \\ = & 8 \times 6^2 + 4 \\ = & 8 \times 36 + 4 \\ = & 288 + 4 \\ = & 292 \end{aligned}$$

leftmost group
exponent inside
division inside
the group
exponent
multiplication

Make the Target Number

Group work. You are given 2, 3, 5, and 8.

Use each number once, with $+$, $-$, $\times$, $\div$ and grouping symbols, to make 48.

Stuck? These numbers are small, so adding them all is not enough. Which two numbers multiply to 48?

Two Ways to Hit 48

Pairs that multiply to 48:

1. 2×24

2. 3×16

3. 4×12

4. 6×8

Pair (4) works: make 6 from 5, 3, and 2.

$$\begin{aligned} & 8 \times (5 + 3 - 2) \\ &= 8 \times 6 = 48 \end{aligned}$$

$$\begin{aligned} & (8 - 2) \times (5 + 3) \\ &= 6 \times 8 = 48 \end{aligned}$$

Every expression you test makes you apply the order of operations again.

One Card, Many Puzzles



Build the target 4 from the five edge numbers. Each number may be used only once.

- **Easy.** Use two or more.
- **Medium.** Use three or more.
- **Hard.** Use four or more.
- **Monstrous.** Use all five.

Can you do it with exactly two numbers? Three? Four? Five?

One Question to Leave With

**Exponents sit above
multiplication in the
order of operations.
What does that buy us?**

An exponent is shorthand for a whole run of multiplications. Doing it first keeps

2×3^2 meaning 2×9 , not $(2 \times 3)^2$.